Week 3: Linux Logs, Sysmon, auditd and Snort Setup

Theory Notes, Developing Basic Understanding in SOC

Question 7. What is MITRE ATT&CK T1046 (Network Service Discovery)? How can Snort rules be mapped to MITRE techniques? Provide an example.

What is T1046

MITRE ATT&CK is a public knowledge base that lists the tactics and techniques attackers have used in real attacks. T1046 is the technique called Network Service Discovery, and it falls under the Discovery tactic. It used to be named Network Service Scanning, and the name was changed in the framework.

- In this technique, the attacker scans other hosts to find out which ports and services are open.
- The goal is to find a weak service they can attack, or a machine they can move to.
- Tools such as Nmap and Masscan are commonly used. A typical example is an Nmap SYN scan, run as `nmap -sS <target>`.

How it can be detected

- On the network, I would look for many SYN packets going to many different ports from one source in a short time. Snort can catch this.
- On the host, Sysmon Event ID 3 can show a process making a large number of outbound connections.
- In firewall logs, one source hitting many ports or many hosts with a lot of denied connections is a warning sign.

Mapping Snort rules to MITRE techniques

Snort does not have a special field only for MITRE, but the mapping can be added inside the rule itself, so that every alert carries it with it.

Rule option	How it helps
msg	Starting the message with the technique ID, such as MITRE T1046, makes it visible in the alert
metadata	A tag like `metadata:mitre_technique T1046;` can be read by a SIEM
reference	Adds the ATT&CK link, for example `reference:url,attack.mitre.org/techniques/T1046/;`
classtype	Gives the category, such as `attempted-recon` for discovery activity

When these alerts reach a SIEM, the alerts can be grouped by technique, which helps to see how much of ATT&CK my rules cover.

Example

```
alert tcp $EXTERNAL_NET any -> $HOME_NET any (msg:"MITRE T1046 - Nmap SYN Port Scan
Detected"; flags:S; threshold:type threshold, track by_src, count 20, seconds 1;
classtype:attempted-recon; metadata:mitre_technique T1046;
reference:url,attack.mitre.org/techniques/T1046/; sid:1000004; rev:1;)
```

- The rule matches SYN packets, because that is how an Nmap SYN scan starts.
- It alerts only when one source sends 20 SYN packets in one second, which is typical scanning behaviour.
- The metadata and reference tags connect the alert directly to T1046.